CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 10: **Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a survey of 45 students, 18 chose cricket. In a pie chart, the central angle of the cricket sector (in degrees) is: 1

(a) 154 **(b)** 54 **(c)** 288 **(d)** 144

2. ₹ 3300 is divided among A, B and C in the ratio 2:4:5. B's share (in ₹) is: 1

(a) 1200 **(b)** 1100 **(c)** 1500 **(d)** 600

3. A car moving at a constant speed covers 180 km in 4 hours. In 10 hours it will cover (km): 1

(a) 450 **(b)** 460 **(c)** 1800 **(d)** 190

4. Which of the following is the greatest ratio? 1

(a) 3:7 **(b)** 2:5 **(c)** 5:12 **(d)** 4:9

5. If A:B = 1:2 and B:C = 3:4, then A:B:C is: 1

(a) 1:2:4 **(b)** 3:8:6 **(c)** 1:5:4 **(d)** 3:6:8

6. 16 men can do a piece of work in 15 days. 24 men will do it in (days): 1

(a) 13 **(b)** 16 **(c)** 10 **(d)** 240

7. On a map the scale is 1 cm = 18 km. Two places 144 km apart are shown on the map at a distance (in cm) of: 1

(a) 8 **(b)** 126 **(c)** 9 **(d)** 2592

8. In a survey of 120 students, 35 chose cricket. In a pie chart, the central angle of the cricket sector (in degrees) is: 1

(a) 210 **(b)** 115 **(c)** 105 **(d)** 95

9. ₹ 2800 is divided among A, B and C in the ratio 3:5:6. B's share (in ₹) is: 1

(a) 1000 **(b)** 934 **(c)** 1200 **(d)** 600

10. A car moving at a constant speed covers 264 km in 6 hours. In 9 hours it will cover (km): 1

(a) 396 **(b)** 2376 **(c)** 273 **(d)** 406

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. x and y are inversely proportional. When $x = 9$, $y = 16$. Find y when $x = 12$. 2

12. y is directly proportional to x. If $y = 48$ when $x = 6$, find the constant of proportionality and y when $x = 9$. 2

13. Which ratio is greater: 11:12 or 9:10? 2

14. x and y are inversely proportional. When $x = 15$, $y = 8$. Find y when $x = 6$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. On a map of scale 1 cm = 5 km, a rectangular park measures 9 cm by 4 cm. Find the actual length, breadth and perimeter of the park. 3

16. A family's monthly expenditure: food 40%, rent 15%, education 10%, others 35%. Find the central angle of each sector of a pie chart. 3

17. A school bus travelling at 40 km/h takes 18 minutes to reach the school. At what speed must it travel to reach in 15 minutes? 3

OR

On a map of scale 1 cm = 4 km, a rectangular park measures 8 cm by 3 cm. Find the actual length, breadth and perimeter of the park. 3

18. A family's monthly expenditure: food 25%, rent 30%, education 25%, others 20%. Find the central angle of each sector of a pie chart. 3

OR

A school bus travelling at 45 km/h takes 16 minutes to reach the school. At what speed must it travel to reach in 12 minutes? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. ₹ 2040 is divided among A, B and C such that A:B = 1:2 and B:C = 3:4. Find the share of each. 5

OR

The marks of a student are: Wheat 14, Rice 10, Maize 8, Bajra 6, Pulses 2. Total 40. Find the central angle of every sector of a pie chart and the percentage of the total taken by Wheat. 5

20. ₹ 1170 is divided among A, B and C such that A:B = 2:5 and B:C = 5:6. Find the share of each. 5

OR

The marks of a student are: Tea 6, Coffee 5, Milk 10, Juice 5, Water 4. Total 30. Find the central angle of every sector of a pie chart and the percentage of the total taken by Tea. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND